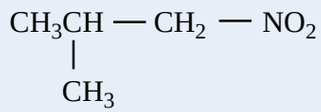
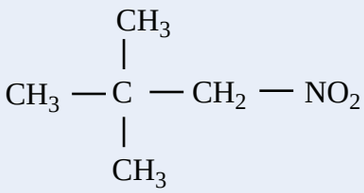
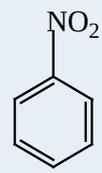
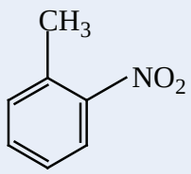
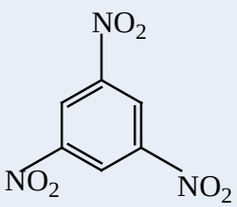
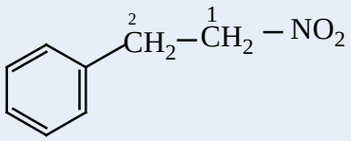
Nitroalkanes are represented by the formula, R-NO_2 where R is an alkyl group (C_nH_{2n+1}-). Nitroalkanes are further classified into primary, secondary, tertiary nitroalkanes on the basis of type of carbon atom to which the nitro (-NO_2) group is attached.

13.1.2 Nomenclature of nitroalkanes

In the IUPAC nomenclature, the nitroalkanes are named by adding prefix nitro before the name of alkane, the position of the nitro group is indicated by number.

Compound (common name, Structural formula, IUPAC Name)	IUPAC Name			
	Prefix with position number	Root used	Primary suffix	Secondary Suffix
 2- Methyl – 1- nitropropane	2- Methyl – 1- nitro	prop	ane	–
 2,2 – dimethyl – 1- nitro propane	2,2 – dimethyl – 1- nitro	prop	ane	–
 Nitrobenzene	nitro	benzene	–	–
 2-nitro -1-methyl benzene	2-nitro-1-methyl	benzene	–	–
 1,3,5 – trinitrobenzene	1,3,5 – trinitro	benzene	–	–
 2 – phenyl – 1- nitroethane	2 – phenyl – 1- nitro	eth	ane	

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